INSTITUTION NAME: Westview University / Central High School

EXAMINATION: End-of-Semester Finals – Spring 2026

Field	Student Input
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Q1. (10 Marks)

Define *data analysis*. Explain the steps involved in a typical data analysis pipeline.

Answer:
Data analysis is the process of inspecting, cleaning, transforming, and modeling data to extract useful insights and support decision-making.

Steps:

1. Data Collection – Gathering raw data from sources
2. Data Cleaning – Handling missing values, removing noise
3. Data Transformation – Normalization, encoding
4. Data Analysis – Applying statistical or ML techniques
5. Data Visualization – Representing insights (graphs, charts)
6. Interpretation – Drawing conclusions

Q2. (10 Marks)

Differentiate between *descriptive* and *predictive* analytics with examples.

Answer:

- **Descriptive Analytics:** Focuses on summarizing past data
  - Example: Monthly sales report
- **Predictive Analytics:** Uses models to forecast future outcomes
  - Example: Predicting next month's sales using regression

Key difference: descriptive = "what happened", predictive = "what will happen".

Q3. (10 Marks)

What is *correlation*? Explain positive, negative, and zero correlation.

Answer:
Correlation measures the strength and direction of the relationship between two variables.

- Positive correlation: both variables increase together
- Negative correlation: one increases while the other decreases
- Zero correlation: no relationship

Example:

- Positive: Study time vs marks

- Negative: Price vs demand

Q4. (10 Marks – Diagram Based)

Explain a *basic data analysis workflow* with a diagram.

Answer:

Raw Data

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Data Cleaning

↓

Data Processing

↓

Analysis (Stats/ML)

↓

Visualization

↓

Insights & Decisions

Explanation:

- Raw data is first cleaned to remove errors
- Processed into usable format
- Analytical techniques are applied
- Results are visualized
- Final insights guide decision-making

End of Paper